

RAPA ITI, French Polynesia

WMO: 919580

Lat: 27.618S

Lon: 144.335W

Elev: 2

StdP: 101.30

Time Zone: -10.00 (W10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.2	14.0	7.1	6.3	16.7	8.0	6.7	17.1	12.4	18.4	11.2	19.0	1.3	200	0.355

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	3.5	28.0	24.4	27.4	24.1	26.8	23.9	25.1	27.1	24.7	26.6	24.3	26.3	4.1	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.4	19.4	26.5	24.0	18.9	26.2	23.7	18.5	25.9	76.3	27.1	74.9	26.8	73.3	26.4	27.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.2	10.1	9.1	11.4	29.0	0.7	0.7	10.9	29.5	10.5	29.9	10.1	30.2	9.6	30.7
			9.9	25.7	0.6	0.6	9.5	26.1	9.1	26.4	8.7	26.7	8.3	27.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	21.0	23.6	24.4	23.9	22.4	20.9	19.5	18.6	18.1	18.3	19.2	20.7	22.0	(6)
	DBStd	2.70	1.47	1.37	1.59	1.59	1.66	1.67	1.67	1.70	1.71	1.53	1.70	1.68	(7)
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD18.3	82	0	0	0	0	1	7	18	25	21	8	2	0	(9)
	CDD10.0	3998	422	403	432	372	337	284	266	252	250	285	322	373	(10)
	CDD18.3	1040	163	170	174	122	79	42	26	19	21	35	74	114	(11)
	CDH23.3	3243	689	966	873	309	69	6	0	1	2	5	71	252	(12)
(13)	CDH26.7	148	23	59	56	7	0	0	0	0	0	0	0	2	(13)
(14)	Wind	WSAvg	4.2	4.2	4.2	4.1	4.5	4.0	4.1	4.1	4.2	4.0	4.0	4.3	(14)
(15) Precipitation	PrecAvg	2611	240	188	258	244	230	219	239	228	174	175	175	226	(15)
	PrecMax	3772	696	487	699	619	578	412	637	567	597	521	592	736	(16)
	PrecMin	1577	25	12	17	48	52	40	93	113	31	5	31	18	(17)
	PrecStd	508	145	110	140	138	133	86	115	109	106	113	142	142	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.1	28.8	28.7	27.6	25.6	24.1	23.1	23.1	23.6	24.0	25.8	27.0	(19)
		MCWB	24.2	24.6	24.9	23.9	22.4	21.5	20.5	20.4	20.6	21.1	22.4	23.6	(20)
	2%	DB	27.4	28.0	28.0	26.5	24.7	23.3	22.5	22.2	22.7	23.2	24.9	26.2	(21)
		MCWB	24.0	24.4	24.4	23.4	22.2	20.9	20.1	19.9	20.1	20.6	21.9	23.3	(22)
	5%	DB	26.7	27.5	27.3	25.7	24.1	22.7	22.0	21.7	22.0	22.6	24.1	25.5	(23)
		MCWB	23.8	24.2	24.2	23.0	21.8	20.4	19.7	19.3	19.6	20.0	21.6	23.0	(24)
	10%	DB	26.1	26.8	26.7	25.1	23.6	22.2	21.4	21.1	21.3	21.9	23.5	24.7	(25)
		MCWB	23.5	23.9	23.9	22.5	21.2	19.8	19.1	18.6	18.8	19.3	21.0	22.5	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.1	25.6	25.6	24.9	23.6	22.4	21.5	21.3	21.4	22.1	23.3	24.5	(27)
		MCDB	26.9	27.7	27.9	26.6	24.6	23.5	22.5	22.4	22.8	23.4	24.8	26.1	(28)
	2%	WB	24.6	25.1	25.1	24.2	23.0	21.7	20.8	20.5	20.7	21.3	22.7	23.9	(29)
		MCDB	26.4	27.1	27.2	25.9	24.2	22.9	22.0	21.9	22.2	22.8	24.1	25.5	(30)
	5%	WB	24.2	24.7	24.7	23.6	22.4	21.0	20.2	19.8	20.1	20.7	22.1	23.4	(31)
		MCDB	26.0	26.7	26.8	25.3	23.8	22.4	21.6	21.3	21.7	22.2	23.7	25.0	(32)
	10%	WB	23.9	24.3	24.2	23.0	21.7	20.3	19.5	19.1	19.2	19.9	21.5	22.9	(33)
		MCDB	25.7	26.4	26.3	24.7	23.2	21.9	21.2	20.7	21.0	21.4	23.1	24.5	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	3.4	3.5	3.8	3.5	3.7	3.9	3.9	4.0	3.8	3.9	3.7	3.4	(35)
		MCDBR	4.4	4.6	4.6	4.5	4.0	4.2	4.0	4.2	4.3	4.5	4.4	4.1	(36)
	5% WB	MCWBR	2.4	2.4	2.6	2.9	2.7	3.0	3.1	3.3	3.1	3.1	2.7	2.4	(37)
		MCWBR	3.7	3.9	4.1	3.7	3.3	3.2	3.3	3.7	3.9	3.6	3.7	3.4	(38)
(40) Clear-Sky Solar Irradiance	taub		0.369	0.372	0.366	0.367	0.355	0.351	0.347	0.347	0.355	0.357	0.355	0.365	(40)
		taud	2.623	2.618	2.644	2.627	2.632	2.623	2.623	2.612	2.573	2.566	2.605	2.608	(41)
	Ebn at Noon		973	954	930	881	847	827	847	888	925	957	980	979	(42)
		Edh at Noon	102	100	92	86	78	75	78	86	97	104	103	104	(43)
(44) All-Sky Solar Radiation	RadAvg		6.52	6.07	5.10	3.99	3.12	2.60	2.91	3.65	4.65	5.67	6.39	6.27	(44)
	RadStd		0.71	0.59	0.57	0.27	0.26	0.19	0.27	0.26	0.33	0.34	0.54	0.51	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page